Project 4 : Laser Tag

Introduction:

PlutoX2 drone can be provided with laser detection capability using a laser tag. This can be used in many applications, we will be working on one of them here.

Challenge:

Build a drone to takeoff, whenever it will sense a laser ray falls on it .

Add-On used:

A laser tag uses a refractive dome canopy that collects incoming laser beams from multiple directions and bends them toward a common focal point. A light sensor placed at this focal point detects the concentrated optical signal. This design improves detection sensitivity and provides wide-angle laser reception

Connection:

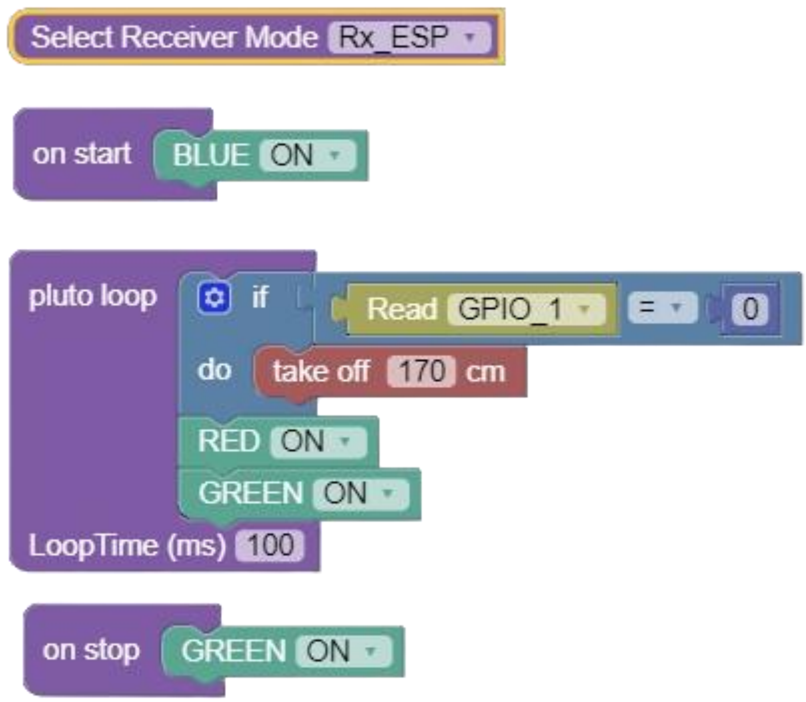
Connect the Laser tag on the RC port on Primus V5/X2 board

Primus V5/X2 pin	Peripherals
VCC on RC port	VCC on Laser tag
GND on RC port	GND on Laser tag
Signal on RC port	A0 on Laser tag

Approaching the Problem:

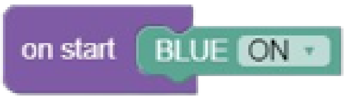
Laser tag checks whether any laser is applied on the drone. When it is, the laser tag output turns LOW and remains until the laser light is turned off. So, whenever laser light falls on the drone, it initiates take off.

Solution:



Explanation of Solution:

On Start Loop:
Blue LED turns on. This is for very small amount of time.



Pluto Loop:
We constantly evaluate the input of Laser tag's sensor. It gives LOW (0), if laser light falls on it. When this happens, drone takes off and flies according to user command. While in background the Yellow light is turned on, indicating the drone is present in developer mode.

On stop Loop: Timer1 resets.

Fig : Pluto with Laser-tag Add-on

Try further!

- Add a buzzer, to know when it crosses a laserlight.
- Make a drone locate within a region guarded invisible walls made up of laser lights.